Kawa Atapour

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Professional summary

Ph.D. student in Electrical and Computer Engineering at Concordia University with research and industrial experience in Machine Learning, Deep Learning, Large Language Models, Federated Learning, Reinforcement Learning, and Explainable AI. Experienced in developing AI solutions for multimodal sensor data analysis, intelligent monitoring systems, and trustworthy AI. Author of multiple peer-reviewed publications in machine learning and distributed AI. Seeking to apply advanced AI and data analytics skills to real-world challenges in the energy sector, including predictive modeling, optimization, intelligent monitoring, and autonomous decision-making.

AI RESEARCH & INDUSTRY EXPERIENCE

AI Research Intern Oct 2025 – Mar 2026
CARE-GAIA: Ambient Assisted Living for People with Dementia, Novatek International, Montreal, Canada
Development of machine learning and large language model (LLM)-based approaches for multimodal sensor data analysis, activity recognition, and intelligent monitoring in smart-home environments for people living with dementia.

AI Research Intern Mar 2026 – Present
CARE-GAIA: Explainable AI for Intelligent Healthcare Systems, Novatek International, Montreal, Canada
Research and development of explainable and reasoning-enhanced AI methods for multimodal human activity recognition, focusing on transparency, interpretability, and trustworthiness in intelligent healthcare systems.

Research Intern Jun 2026 – Present
From Agentic AI to Autonomy for Trustworthy Human-AI Teaming, Defence Research and Development Canada (DRDC), Canada
Research on trustworthy AI, autonomous intelligent agents, predictive intelligence, and human-AI teaming, with emphasis on reasoning, adaptation, and reliable decision-making in dynamic environments.

TEACHING EXPERIENCE

Teaching Assistant 2025 – Present
Discrete Signal Processing (ELEC 342), Concordia University, Montreal, Canada Assisted in the delivery of undergraduate signal processing courses, supported laboratory sessions and assignments, and provided guidance on signal analysis, transforms, filtering, and practical implementation using MATLAB and Python.

Teaching Assistant 2020 – 2021
Digital Image Processing, Tarbiat Modares University, Tehran, Iran Supported laboratory sessions and tutorials covering image enhancement, filtering, segmentation, feature extraction, and machine learning techniques for image analysis.

Teaching Assistant 2020 – 2021
Neural Networks, Tarbiat Modares University, Tehran, Iran Assisted students with neural network fundamentals, deep learning concepts, optimization methods, and practical implementation of machine learning models using Python.

Teaching Assistant Feb 2020 – Jan 2021
Advanced Digital Signal Processing Laboratory, Tarbiat Modares University, Tehran, Iran Delivered tutorials on digital signal processing, image processing, and machine learning concepts, and guided students in implementing algorithms using Python and MATLAB.

EDUCATION

Doctor of Philosophy in Electrical and Computer Engineering **May 2025 – Present**
Concordia University, Montreal, Canada *GPA: 4/4*

Master of Science in Electrical Engineering, Communication Systems **Oct 2018 – Oct 2021**
Tarbiat Modares University, Tehran, Iran *Tehran, Iran*
- Thesis Title: Traffic-Based Multi-Tenant Cross-Slice Resource Orchestration in Future Wireless Networks

Bachelor of Science in Electrical Engineering **Oct 2013 – Oct 2017**
Bu-Ali Sina University of Hamedan *Hamedan, Iran*
- Thesis Title: Introduction to image processing and segmentation for Multiple Sclerosis diagnosis

PUBLICATIONS

- **[JR-1]S. K. Atapour**, et al., "ECLIPSE: Embedding-Centered Foundation-Model Federated Knowledge Transfer for Data-Free Edge Intelligence," Submitted to *IEEE Transactions on Reliability, Trustworthy AI Special Issue*, 2026. (**First Author**)
- **[JR-2]** A. Alinezhadi, S. M. Sheikholeslami, **S. K. Atapour**, J. Abouei, and K. N. Plataniotis, "Intelligent Privacy-Preserving Demand Response for Green Data Centers," *Electric Power Systems Research*, vol. 221, Art. no. 109394, Aug. 2023. (**Co-author**)
- **[C-1]** S. J. Seyedmohammadi, **S. K. Atapour**, J. Abouei, and A. Mohammadi, "KNFu: Effective Knowledge Fusion," *27th International Conference on Information Fusion* (*FUSION*), 2024. (**Co-author**)
- **[C-2] S. K. Atapour**, S. J. Seyedmohammadi, J. Abouei, and A. Mohammadi, "Multi-Model Federated Learning Optimization Based on Multi-Agent Reinforcement Learning," *IEEE CAMSAP Workshop*, 2023. (**First Author**)
- **[C-3]** E. Rezaei, S. J. Seyedmohammadi, **S. K. Atapour**, J. Abouei, et al., "CoCo-POI: Collaborative Contrastive Learning for Point-of-Interest Recommendation," *IEEE International Conference on Human-Machine Systems* (*ICHMS*), 2025. (**Co-author**)
- **[AUT-1] S. K. Atapour**, et al., "FedD2S: Personalized Data-Free Federated Knowledge Distillation," *arXiv preprint*, 2024. (**First Author**)
- **[AUT-2] S. K. Atapour**, et al., "Leveraging Foundation Models for Efficient Federated Learning in Resource-Restricted Edge Networks," *arXiv preprint*, 2024. (**First Author**)
- **[AUT-4] S. K. Atapour**, et al., "JEPA-Based Predictive Representation Learning for World Models and Autonomous AI Systems," Manuscript in preparation for submission to *IEEE Signal Processing Letters*. (**First Author**)

RESEARCH INTERESTS

- Machine Learning and Deep Learning
- Foundation Models and Large Language Models (LLMs)
- Predictive Representation Learning and World Models
- Explainable and Trustworthy AI
- Federated Learning and Distributed Intelligence
- Reinforcement Learning and Autonomous Systems
- Multimodal Learning and Sensor Data Analytics

TECHNICAL SKILLS

- Programming Languages: Python, MATLAB, C, C++, R
- Machine Learning & AI: Deep Learning, Reinforcement Learning, Federated Learning, Knowledge Distillation, Self-Supervised Learning, Foundation Models, Large Language Models (LLMs), Explainable AI
- Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib
- Data Analytics: Statistical Analysis, Predictive Modeling, Time-Series Analysis, Multimodal Data Analysis
- Signal Processing: Digital Signal Processing, Image Processing, Sensor Data Analytics, Feature Extraction
- Platforms & Hardware: Linux, Raspberry Pi, Arduino

HONORS & AWARDS

- **Tuition Award of Excellence for International Students**, Concordia University, Montreal, QC, Canada, 2025.
- **Mitacs Accelerate Research Internship Award**, CARE-GAIA Project, Novatek International, Montreal, QC, Canada, 2025–2026.
- **Mitacs Accelerate Research Internship Award**, CARE-GAIA Project, Novatek International, Montreal, QC, Canada, 2026.